



AI-Powered Collections Strategy



Leveraging Agentic AI for Scalable, Fair, and
Effective Debt Management at Geldium



How the System Works

- **Inputs:**

- 1- Customer demographic information, income, credit score, and employment status are collected through internal financial systems and credit bureau data.
- 2- Real-time payment activity, credit utilization, loan balances, and repayment history are continuously monitored to identify delinquency risk patterns.

- **Decision Logic:**

- 1- The system combines predictive model outputs with predefined business rules to classify customers into low, medium, or high-risk categories.
- 2- AI-driven risk scores help determine the most appropriate intervention strategy based on customer behavior and financial risk indicators.

- **Actions:**

- 1- Automated interventions include personalized SMS reminders, email notifications, repayment assistance offers, and escalation workflows for high-risk accounts.
- 2- Customers showing signs of financial hardship may receive flexible repayment plans or proactive support options.

- **Learning:**

- 1- The system continuously tracks repayment outcomes, customer engagement, and intervention effectiveness to improve future decision-making.
- 2- AI models are regularly updated using new customer behavior data to enhance prediction accuracy, fairness, and operational performance.

Role of Agentic AI

<u>Autonomous AI Activities</u>	<u>Human Oversight Activities</u>
Real-time customer risk scoring and delinquency prediction	Review high-risk or sensitive customer cases
Automated SMS, email, and payment reminder generation	Approve hardship assistance or repayment restructuring
Personalized outreach timing and communication strategies	Monitor fairness, bias, and regulatory compliance
Continuous tracking of customer repayment behavior	Handle customer disputes and escalation decisions
Adaptive learning based on customer response patterns	Conduct periodic audits and model performance reviews

=> The agentic AI system automates repetitive operational tasks while maintaining human oversight for high-impact financial decisions to ensure fairness, transparency, and responsible customer treatment.

Responsible AI Guardrails

- Bias detection and fairness monitoring across all customer segments.
- Transparent and explainable AI-driven risk assessment processes.
- Human review for critical, high-impact, or disputed decisions.
- Regulatory compliance with GDPR, ECOA, FCA, and financial governance standards.
- Secure storage and responsible handling of sensitive customer financial data.
- Continuous model monitoring, auditing, and performance validation.
- Ethical AI governance to ensure accountability and responsible automation.
- Customer-centric decision-making focused on fairness, trust, and transparency.
- Real-time compliance checks integrated into automated workflows.
- Regular retraining of AI models using updated and representative datasets.
- Prevention of proxy discrimination from sensitive variables such as location or income group.
- Detailed audit trails for tracking AI-generated decisions and interventions.
- Escalation mechanisms for unusual or high-risk customer cases.
- Periodic regulatory and ethical review of AI system performance.
- Controlled access management to prevent unauthorized data usage.
- Explainable communication of AI decisions to customers and stakeholders.

Expected Business Impact

Business KPIs

- Reduction in customer delinquency and missed payment rates.
- Improved repayment recovery and collections efficiency.
- Faster identification of high-risk customer segments.
- Lower operational costs through automated outreach workflows.
- Increased scalability of collections operations across large customer bases.
- Improved accuracy in prioritizing customer intervention strategies.

Customer Outcomes

- More personalized and proactive customer engagement.
- Fairer and more transparent financial decision-making.
- Improved customer trust and repayment experience.
- Faster access to hardship assistance and support programs.
- Reduced customer stress through timely communication and flexible repayment options.
- Enhanced customer satisfaction through responsible AI-driven interactions.

Conclusion & Next Steps

- AI-powered collections systems can improve delinquency prediction, operational efficiency, and customer engagement.
- Agentic AI enables adaptive and personalized intervention strategies while maintaining scalability.
- Human oversight, fairness monitoring, and regulatory compliance remain essential for responsible AI deployment.
- Continuous learning and model refinement will help improve prediction accuracy and long-term business performance.
- The proposed framework supports Geldium's goal of building a smarter, fairer, and more customer-centric collections strategy.

“The proposed AI-driven collections framework balances automation, operational efficiency, customer fairness, and responsible financial decision-making.”